Data Handling: Import, Cleaning and Visualisation

Exercise 5: Data Preparation and Data Analysis with R

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Exercises[1]

The Canvas page contains csv files with data needed to solve the following exercises. Make sure to store them in your `r_course/data/` folder. Do not forget to load `tidyverse` before you get started!

Exercise A

Read `stocks.csv` into RStudio (store it in variable `stocks`). The dataset is currently in long format. Use `pivot_wider()` to transform it into wide format (separate columns for different years). Store the resulting `tibble/data.frame` in a variable called `stocks2`. Then, run the following code.

```
pivot_longer(stocks2, `2015`:`2016`, names_to = "year", values_to = "return") == stocks
```

Why is the result not identical to the original dataset (`stocks`)? In other words, why are `pivot_longer()` and `pivot_wider()` not perfectly symmetrical? (Hint: look at the variable types and think about column names.)

Exercise B

Read `countries.csv` into RStudio (store it in variable `countries`). Then, run the following code.

```
pivot_longer(countries, 1999, 2000, names_to = "year", values_to = "cases")
```

Why does this code fail? How can you fix the problem?

Exercise C

Read `people.csv` into RStudio (store it in variable `people`). Try to use `pivot_wider` on `people`. Why does it fail? How could you add a new column to fix the problem?

Exercise D

Read `preg.csv` into RStudio (store it in variable `preg`). Tidy `preg`. Do you need to use `pivot_wider` or `pivot_longer`? What are the variables?

Exercise E

Execute the following commands, which will create a simulated dataset representing four countries, the number of Covid cases in each of these countries in 2020 and 2021, as well as the public spending dedicated to fighting Covid in 2020 and 2021.

[1] Adopted from https://r4ds.had.co.nz/tidy-data.html.

```
df <- data.frame("Country" = c("US", "DE", "CH", "FR"),
                 "Covid_2020" = runif(4, 0, 1),
                 "AntiCovidExpenses_2020" = runif(4, 500, 1000))
df <- cbind(df,
            "Covid_2021" = df$Covid_2020 + rnorm(4, 0, 0.1),
            "AntiCovidExpenses_2021" = df$AntiCovidExpenses_2020 + rnorm(4, 0, 100))
```

You would like to investigate the correlation between public spending and the number of Covid cases across years and across countries. How do you proceed? Is this dataset long or wide? (Note: these data are simulated and are in no way representative of the reality.)

Additional Exercises (advanced)

Exercise F

You want to investigate the relationship between GDP per capita and CO2 emissions per capita across countries. On Canvas, download the dataset `co2gdp.csv`. This dataset has been downloaded from the website "Our World in Data" (https://ourworldindata.org/grapher/co2-emissions-vs-gdp). Investigate the relationship between the dependent variable `CO2_percapita` and the independent variable `GDP_percapita` using a linear regression of the following form:

$$\text{CO2\_percapita}_i = \alpha + \beta \text{GDP\_percapita}_i + u_i.$$

Estimate this linear regression by creating your own function `myols` that takes the following three arguments: (x) for the dependent variable, (y) for the independent variable, and *matrix*, which takes `TRUE` if the function estimates the coefficients using matrix algebra, and `FALSE` if the function estimates the coefficients using the sum operators. Compare the coefficients with the `R` built-in function `lm()`. Finally, comment and interpret the coefficients.

Hint 1 (linear regression with sum operator): the slope parameter of the linear regression is expressed as the following formula:

$$\hat{\beta} = \frac{\sum x_i y_i - N\bar{y}\bar{x}}{\sum x_i^2 - N\bar{x}^2},$$

and the constant is given by this formula:

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x},$$

where $\bar{x}$ gives the mean of x and $\bar{y}$ gives the mean of y.

Hint 2 (linear regression with matrix algebra): the formula for the linear regression in matrix algebra is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-\mathbf{1}}\mathbf{X}\mathbf{y},$$

where you find the inverse of a matrix with the command 'solve()', the cross-product of two matrices with `crossprod(X,Y)` $= t(X)\%{*}\%Y$. Also, note that you need a column of 1s in your matrix of covariates to compute the intercept.

(**Note**: this exercise is about programming and implementing the formula of the linear regression. You don't need to understand the mechanics of linear regression for the exam. This will be covered next semester in the course `Data Analytics II`.)

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. The following line of R code will return `"bnn"`.

str_split(“banana”, pattern= “a”)

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. Regarding the data semantics (vocabulary used to describe data) introduced in this course, which of the following statements is correct?

A) Datasets typically contain more variables than values.
B) An observation contains all values measured on the same unit (e.g., a person).
C) A variable contains all values that measure the same underlying attribute across units/observations.
D) Every observation belongs to a value and a variable.

Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. Which of the following statements about ‚tidy‘ data is correct?

A) `tibbles` can only contain tidy data.
B) Datasets in wide format are generally not considered ‚tidy‘.
C) When a csv-file is imported with `read_csv()`, the function issues a warning if the data imported is not tidy.
D) Data stored in csv-format is generally referred to as ‚tidy data‘.